

Separating linearized observability, direct recoverability, and workflow limitations in physics-informed neural network inversion

Fei Yang^{1*}

¹China Institute of Geo-Environment Monitoring, China Geological Survey, Minzu University South Road, Beijing, 100081, China.

Corresponding author(s). E-mail(s): yangf@cigem.cn;

Abstract

Physics-informed neural networks (PINNs) are increasingly used for groundwater inverse problems, but a good hydraulic-head fit does not establish conductivity recovery. We present a controlled diagnostic study of PINN-KLE conductivity inversion for steady two-dimensional groundwater flow, aiming to separate local information content, gradient transfer, round-selection quality, and workflow-limited recovery—where recoverable information exists in the direct reduced map but is not reliably transmitted by the staged workflow.

Gradient audits show that the head-observation loss provides no direct gradient pathway to KLE coefficients; updates are mediated through the PDE-residual branch in the implemented staged workflow, whose direction is weakly aligned with truth. A no-truth round selector identifies the best alternating round in 17/18 cases, yet only 3/18 weak-pass conductivity recovery. The direct FDM-KLE baseline weak-passes 24/24 cases across eight truth realizations ($\sigma_{\log \kappa} = \mathbf{0.17-0.80}$), confirming that the sparse inverse problem contains recoverable low-dimensional information; the failure is therefore workflow-limited relative to the matched direct baseline. Jacobian diagnostics show full linearized local rank for 3/5/10-mode direct maps at $n_{\text{obs}} = \mathbf{200}$ and for 3-mode maps across 50–400 observation points, with conditioning burden increasing with mode count. Local sensitivity at hybrid-recovered coefficients remains comparable to true-point conditioning (condition-number ratio 0.919–1.010, median 0.953, across twelve paired cases), indicating that failures do not arise from loss of local direct-map observability. The paper argues for a more careful diagnostic standard: head fitting, linearized local observability, direct deterministic baselines, and closed-loop conductivity recovery must be reported separately.

Keywords: Groundwater inverse problem, Physics-informed neural networks, Karhunen–Loève expansion, Local observability, Workflow diagnostics

1 Introduction

Groundwater conductivity inversion is a classical ill-posed inverse problem. Sparse hydraulic-head observations provide indirect and smoothed information about subsurface heterogeneity, and the

answer obtained from an inverse model depends strongly on parameterization, prior structure, boundary forcing, and the richness of hydraulic excitation [1–6]. The identifiability of aquifer

parameters from limited head data has been studied extensively: Sun and Yeh [7] introduced coupled inverse-analysis and experimental design concepts, Hill and Tiedeman [8] developed sensitivity-based calibration methods, and Tarantola [9] and Aster et al. [10] provided general inverse-theory frameworks for parameter estimation. Hydraulic tomography further demonstrated that multiple pumping tests can improve subsurface characterization [11–13]. Physics-informed neural networks offer an attractive differentiable framework for combining neural approximators with partial differential equations [14, 15], and they have been applied to subsurface-flow parameter learning and groundwater inverse modeling [16–20]. However, PINNs do not remove the information-content problem that makes groundwater inversion difficult. A neural model can match observed heads while still failing to identify conductivity parameters.

This paper focuses on that evaluation gap. In some recent PINN inverse studies, the reported evidence is dominated by head misfit, PDE residual, or visual agreement. For conductivity inversion studies, these diagnostics are insufficient unless they are connected to coefficient-level recovery. In a PINN-KLE setting, one must ask whether the data term produces gradients with respect to the KLE coefficients, whether alternating training rounds can be selected without using full-field truth, whether recovered coefficients pass an internal consistency check, and whether a direct physics-based inverse baseline can recover coefficients in cases where the tested PINN workflow does not.

We study these questions in a deliberately controlled steady groundwater benchmark. The benchmark is not intended to reproduce all features of a field aquifer. Instead, it is designed to make different failure modes separable. A failure may be structural, meaning that the observations and equations do not determine the parameter. It may be a practical recovery failure under finite noisy data. It may be local uncertainty in a shallow objective valley. Or it may be workflow-limited: a direct baseline can recover the reduced parameters, but the tested PINN-hybrid workflow cannot transmit that information into a conductivity estimate.

The contributions are:

1. a controlled numerical diagnostic study for sparse-head PINN-KLE groundwater inversion;
2. a loss-structure gradient audit showing that the standard head-only data loss does not directly update KLE coefficients;
3. a no-truth round-selection audit that separates choosing the best available alternating round from achieving conductivity recovery;
4. a matched direct FDM-KLE least-squares baseline showing that the main moderate-noise 3-mode failure is consistent with a workflow-limited interpretation in the tested setting;
5. sparse-head Jacobian, Fisher-information, and Gauss-Newton diagnostics showing linearized local observability for the direct map, increasing conditioning burden with mode count, and comparable conditioning at hybrid-recovered coefficients—supporting the interpretation that the workflow failure does not arise from loss of local direct-map observability at the recovered parameters.

The central claim is deliberately narrow: the tested PINN-hybrid workflow can lose coefficient information that is still accessible to a direct deterministic FDM-KLE inverse baseline. The paper does not claim that all PINN inverse formulations fail, nor that the numerical thresholds found here transfer unchanged to real aquifers.

2 Problem setup and parameterization

2.1 Governing equation and boundary conditions

All experiments solve a steady two-dimensional saturated groundwater flow problem on the unit square:

$$\nabla \cdot (K \nabla h) = 0, \quad (x, y) \in (0, 1)^2, \quad K(x, y) > 0. \quad (1)$$

The default boundary conditions impose fixed heads at the left and right boundaries and no-flow conditions at the top and bottom:

$$h|_{x=0} = 1, \quad h|_{x=1} = 0, \quad \mathbf{n} \cdot K \nabla h|_{y=0, y=1} = 0. \quad (2)$$

This source-free steady setup is intentionally simple. It removes pumping, recharge, transient

response, irregular geometry, and conceptual-model uncertainty so that workflow-level failures can be interpreted without field ambiguity. The limitation is equally important: numerical thresholds reported here are benchmark-specific and should not be treated as monitoring-network design rules for real aquifers.

2.2 KLE log-conductivity model

The heterogeneous log-conductivity field is represented with a truncated KLE:

$$\ln K(x, y) = \overline{\ln K} + \sum_{i=1}^M \xi_i \sqrt{\lambda_i} \phi_i(x, y), \quad (3)$$

where $\overline{\ln K}$ is the mean log-conductivity, λ_i and ϕ_i are the eigenvalues and eigenfunctions of the spatial covariance kernel, ξ_i are the dimensionless stochastic coefficients, and M is the truncation order. The coefficient vector $\boldsymbol{\xi} = (\xi_1, \dots, \xi_M)^\top$ is the target reduced-order coefficient vector of the inverse problem. Most matched-noise comparisons use a 3-mode KLE target ($M = 3$). Additional mode-bridge experiments examine $M = 5$ and $M = 10$ settings. The KLE representation follows standard KLE formulations for random fields and flow in heterogeneous media [21–23]. It is used as a controlled reduced parameterization, not as a universal geological model. It lets us ask whether low-dimensional information is present in the data and whether the tested workflow can recover it.

2.3 Synthetic observations and closed-loop validation

Synthetic sparse head observations are sampled from FDM reference solutions and perturbed by controlled Gaussian noise. Recovery quality is evaluated through a closed-loop FDM solve: the estimated conductivity coefficients are inserted back into the FDM forward model, and the resulting hydraulic head is compared with the reference field. The main metrics are:

- closed-loop head RMSE;
- log-conductivity RMSE;
- KLE coefficient correlation;
- weak-pass and strong-pass operational recovery labels (weak-pass: $\log K$ RMSE < 0.15 and

h RMSE < 0.006 ; strong-pass: $\log K$ RMSE < 0.10 and h RMSE < 0.003).

The pass labels are audit thresholds, not physical phase transitions. They are used to summarize whether a run achieved a useful coefficient recovery under the stated benchmark criteria. The closed-loop check uses the same FDM forward family as the reference solution; it is an internal consistency check on the coefficient estimate, not an independent model validation against a different physics solver.

3 Diagnostic methods

3.1 PINN-KLE gradient audit

In the tested staged PINN-KLE formulation, a neural network approximates the head field, while the KLE coefficients are treated as trainable variables. The loss contains head-observation, boundary-condition, and PDE-residual terms. We compute component-wise gradients of these loss terms with respect to the KLE coefficients. This diagnostic asks a simple question: which terms actually provide coefficient-level gradient information?

The gradient audit is not a convergence test. It is a structural diagnostic of the loss pathway. In the tested workflow, the head-observation loss updates only the neural head approximator and provides no direct gradient pathway to $\boldsymbol{\xi}$; therefore, head fitting alone cannot by itself serve as sufficient evidence of conductivity recovery. This diagnostic is workflow-specific and does not describe all PINN formulations—it reflects the staged-freeze design of the tested workflow.

3.2 Hybrid PINN-FDM workflow

The staged PINN-FDM workflow separates head approximation from coefficient recovery:

1. Stage A fits a neural head field using sparse observations and physics constraints.
2. Stage B freezes the sampled head field on a grid and optimizes KLE coefficients through a discrete residual.
3. Stage C performs an FDM forward solve using the recovered coefficients and evaluates closed-loop metrics.

Alternating Stage A/B/C rounds are repeated. Earlier experiments used oracle full-field head RMSE to choose the best round, which is not available in field applications. The current diagnostic uses sparse-observation RMSE as a no-truth selector and audits it against oracle head RMSE only in synthetic mode.

3.3 Direct FDM-KLE least-squares baseline

The direct baseline optimizes the same KLE coefficients against the same sparse head observations using repeated FDM forward solves. It is a deterministic parameter-estimation baseline in the same reduced KLE space. This baseline is not intended to mimic every feature of PEST, EnKF, or other hydrogeologic inversion systems. Its role is sharper: if the direct FDM-KLE baseline recovers the coefficients while the PINN-hybrid workflow fails, then the failure cannot be attributed simply to absence of low-dimensional information in the observations.

3.4 Workflow-diagnostics aggregation

The workflow diagnostics package combines five post-processing audit layers. Each layer addresses a specific failure-mode question. None of the layers requires re-running the forward or inverse model; they aggregate existing training outputs into a diagnostic chain.

Layer 1: Gradient-transfer audit.

For each loss component $\mathcal{L}_k$ (head observation, boundary condition, PDE residual), we record the gradient vector $\nabla_{\xi} \mathcal{L}_k$ at selected training steps. The audit asks: which loss terms produce non-zero gradients with respect to the KLE coefficients? If the data loss provides no direct computational gradient pathway, then head fitting alone cannot drive coefficient updates. We also compute the cosine similarity between the PDE-loss gradient direction and the direction from the current ξ to ξ_{true} to assess gradient alignment.

Layer 2: Selector-versus-recovery audit.

After all alternating rounds are completed, we evaluate two independent criteria: (a) the *selector* quality, defined as the sparse-observation RMSE

of the chosen round relative to the oracle-best round, and (b) the *recovery* quality, defined as the closed-loop logK RMSE of the chosen round. A round can have excellent selector quality (near-zero oracle regret) but poor recovery quality. This separation is essential because field applications must select rounds without access to truth.

Layer 3: Matched direct-baseline gap.

For each PINN-hybrid case, we run a direct FDM-KLE least-squares inversion (Section 3.3) using the same truth, same observation locations, and same noise realization. The gap between the direct baseline’s logK RMSE and the hybrid’s logK RMSE measures workflow-limited information loss. If the direct baseline recovers the coefficients and the hybrid does not, the failure is not attributable to absence of low-dimensional information in the data.

Layer 4: Late-round degradation audit.

We track the per-round Stage A head RMSE, Stage B logK RMSE, and Stage C closed-loop head RMSE across alternating rounds. A transition is classified as *degrading* if the Stage C RMSE worsens from one round to the next. For each degrading transition, we record whether Stage A head RMSE, Stage B logK RMSE, or both rebounded, and compute the correlation between Stage B logK-RMSE change and Stage C head-RMSE change. This identifies whether late degradation is driven by Stage A fitting failure, Stage B coefficient damage, or A/B coupling.

Layer 5: Mode-bridge diagnostic.

We compare hybrid recovery with a perfect-head coefficient recovery reference. The perfect-head reference solves the inverse problem using the full-field FDM head solution as input rather than a neural approximation. The ratio of hybrid logK RMSE to perfect-head logK RMSE quantifies the workflow damage attributable to the neural head approximation versus the increasing conditioning burden of higher mode counts.

3.5 Sparse-head Jacobian, Fisher information, and Gauss-Newton Hessian

For the direct FDM-KLE observation map, we compute a finite-difference sensitivity matrix (the Jacobian) at a reference coefficient vector ξ^* :

$$J_{ij} = \frac{\partial h(\mathbf{x}_i; \xi^*)}{\partial \xi_j}, \quad (4)$$

$$i = 1, \dots, n_{\text{obs}}, \quad j = 1, \dots, M,$$

where $\mathbf{x}_i$ are the sparse observation locations. The finite-difference approximation uses an absolute perturbation $\delta = 10^{-3}$:

$$J_{ij} \approx \frac{h(\mathbf{x}_i; \xi^* + \delta \mathbf{e}_j) - h(\mathbf{x}_i; \xi^* - \delta \mathbf{e}_j)}{2\delta}. \quad (5)$$

From this matrix we form the local Fisher-information approximation under least-squares Gauss-Newton linearization and the Gauss-Newton Hessian approximation [7, 9, 10]:

$$\mathbf{F} = \frac{1}{\sigma^2} \mathbf{J}^\top \mathbf{J}, \quad \mathbf{H}_{\text{GN}} = 2 \mathbf{J}^\top \mathbf{J}, \quad (6)$$

where σ^2 is the observation-noise variance. Under the standard assumption of iid Gaussian observation errors with variance σ^2 , $\mathbf{F}$ corresponds to the Fisher information matrix for the linearized observation model. Because $\mathbf{F}$ and $\mathbf{H}_{\text{GN}}$ share the same rank and their singular-value ratios are determined by $\mathbf{J}$ alone (up to the noise-dependent scaling $1/\sigma^2$), the Jacobian rank and singular-value spectrum suffice to diagnose local observability and conditioning of the direct reduced inverse map. Under the present least-squares Gauss-Newton formulation, scanning the noise level σ only rescales the eigenvalues and does not change the rank or condition number of $\mathbf{F}$. We note that the condition numbers satisfy $\kappa(\mathbf{H}_{\text{GN}}) = \kappa(\mathbf{F})$ and, for the standard spectral condition number, $\kappa(\mathbf{H}_{\text{GN}}) \approx \kappa(\mathbf{J})^2$; the larger HGN values reported in the results reflect this squaring relationship and do not indicate a qualitatively different conditioning regime.

These diagnostics are local and linearization-based: they do not prove global identifiability, uniqueness, or absence of multimodal structure [7, 8]. They do, however, test whether small perturbations in all coefficient directions are detectable

by the sparse-head map at the linearization point. We apply the Jacobian diagnostic at two types of reference points: (a) the true coefficient vector and (b) the hybrid-recovered coefficient vector, to test whether workflow failure coincides with loss of local observability.

4 Results

4.1 Head-observation loss does not directly identify KLE coefficients

The gradient audit shows that in the tested staged workflow, the head-observation loss provides no direct gradient pathway to KLE coefficients both before and after Stage A. In the representative diagnostic case, the initial PDE-loss coefficient-gradient norm is 4.84×10^3 , while the after-Stage-A PDE-loss coefficient-gradient norm falls to 1.68. The PDE gradient direction is only weakly aligned with the direction to the true coefficients: the cosine is 0.200 initially and 0.309 after Stage A. The first actual coefficient-update direction has cosine 0.056 with the direction to truth.

This explains why head fitting alone is a weak inverse-success diagnostic in the tested PINN-KLE formulation. Coefficient information reaches ξ through the PDE-residual branch in the implemented staged workflow, not directly through the sparse head data term. This is consistent with broader PINN evidence that optimization pathologies and gradient imbalance can strongly affect whether physics-informed losses transmit useful parameter information [24, 25] (Figure 1).

4.2 No-truth round selection is not the same as recovery

The observation-RMSE selector performs well in the current medium-budget pilot aggregate, which spans five noise levels ($\sigma = 0.005, 0.01, 0.015, 0.02, 0.05$) and therefore extends beyond the compact matrix's three-level scan. Across 18 cases, all 18 are hard-pass selector cases. However, only 1 of these 18 cases passes the conductivity-recovery criterion. The median hybrid logK RMSE increases from 0.153 at noise $\sigma = 0.005$ to 0.178, 0.236, 0.318, and 1.133 at noise levels $\sigma = 0.01, 0.015, 0.02$, and 0.05, respectively.

The conclusion is not that the selector fails. Rather, it correctly answers a limited operational

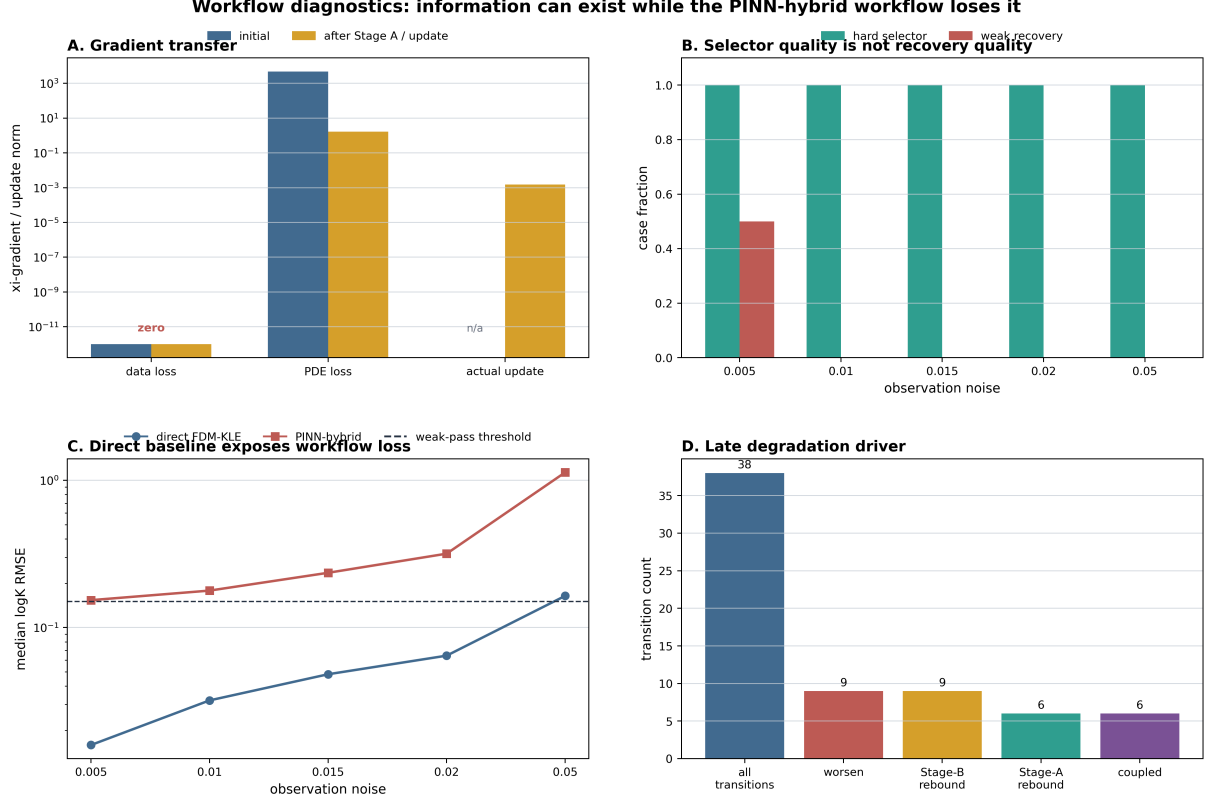


Fig. 1 Workflow diagnostics for the tested PINN-KLE inverse procedure. Panel A summarizes the coefficient-gradient transfer audit. Panel B separates no-truth selector quality from conductivity recovery (fractions computed across noise groups of unequal case counts; see Section 4.2). Panel C compares median logK RMSE for direct FDM-KLE and PINN-hybrid cases. Panel D summarizes late-round degradation drivers; bars are not mutually exclusive (total transitions = 38, of which a subset co-occur).

question: which available alternating round is best under a no-truth sparse-observation criterion? That question is distinct from whether the best available round contains a useful conductivity estimate.

4.3 Direct FDM-KLE exposes workflow-limited recovery loss

The direct FDM-KLE comparison is the central workflow diagnostic. We present three progressively controlled experimental phases: an initial medium-budget pilot aggregate, a compact multi-truth matrix, and a targeted three-seed audit.

Pilot aggregate.

Across the initial matched aggregate, direct FDM-KLE weak-passes 15/18 cases, while PINN-hybrid weak-passes 1/18. The median logK RMSE of the

direct baseline remains below or near the weak-pass threshold through moderate noise, whereas the PINN-hybrid median logK RMSE is already above threshold at noise $\sigma = 0.005$ and grows with noise.

At noise $\sigma = 0.01$ and 0.015 , the direct baseline weak-passes all matched fine transition cases, while the medium-budget PINN-hybrid workflow weak-passes none. At noise $\sigma = 0.02$, the direct baseline again weak-passes all matched cases in the medium-budget comparison, while the PINN-hybrid workflow fails. These results support the workflow-limited interpretation: the sparse direct inverse problem still contains recoverable low-dimensional information, but the tested PINN-hybrid workflow does not reliably transmit it into recovered coefficients.

Compact multi-truth matrix.

As a controlled multi-truth check, we ran direct FDM-KLE on eight 3-mode truth realizations spanning $\log K$ standard deviations from 0.17 to 0.80, at $n_{\text{obs}} = 200$ and three noise levels ($\sigma = 0.005, 0.01, 0.02$) using a single observation seed. The direct baseline passed all 24 cases (8×3) under both weak and strong criteria. Median closed-loop $\log K$ RMSE increased with noise, from 0.018 at $\sigma = 0.005$ to 0.036 at $\sigma = 0.01$ and 0.072 at $\sigma = 0.02$, remaining well below the weak-pass threshold ($\log K$ RMSE < 0.15 , h RMSE < 0.006). In the most challenging display truth (8, $\sigma_{\log K} = 0.80$), $\log K$ RMSE reached 0.089 at $\sigma = 0.02$ but still satisfied the weak-pass criterion. The full 8-truth results are provided in Online Resource 1. By contrast, the five additional truths are evaluated as FDM-only baselines; matched PINN-hybrid comparisons remain with the three original display truths (1, 2, 3) for which seed-audit hybrid data already exist.

The matched PINN-hybrid expansion was then run for the more favorable $n_{\text{obs}} = 200$ subset of the same three truths and three noise levels. The contrast is sharp. In the matched $n_{\text{obs}} = 200$ cases, direct FDM-KLE passes all 9 cases under both weak and strong criteria. PINN-hybrid produces hard-pass no-truth selector outcomes in all 9 cases, with zero oracle head-RMSE regret, but weak-passes only 2/9 and strong-passes 0/9. By noise, PINN-hybrid weak-passes 1/3 at $\sigma = 0.005$, 1/3 at $\sigma = 0.01$, and 0/3 at $\sigma = 0.02$; the corresponding median $\log K$ RMSE values are 0.590, 0.626, and 0.777. By truth, the workflow is near the weak-pass boundary for truth 1, but fails all tested noise levels for truths 2 and 3. The compact matched matrix therefore reinforces the separation between direct recoverability, round-selection quality, and workflow recovery.

Three-seed audit.

To check whether this contrast is an artifact of a single observation/training seed, we repeated the compact comparison at $n_{\text{obs}} = 200$ for the two bracketing noise levels ($\sigma = 0.005$ and 0.02) using three seeds (202, 303, and 404). Across these 18 seed-audit cases, direct FDM-KLE weak-passes all 18 and strong-passes 15. In the matched PINN-hybrid audit, the selector remains operationally reliable in 17/18 cases with median oracle

head-RMSE regret equal to zero, but conductivity recovery weak-passes only 3/18 and never strong-passes. The truth dependence persists: truth 1 weak-passes 3/6 hybrid cases, whereas truths 2 and 3 weak-pass 0/6 each. This targeted seed audit does not replace a full statistical ensemble, but it removes the most immediate single-seed explanation for the observed workflow-limited gap. Seeds 202, 303, and 404 were selected as a targeted bracket rather than a random draw from a full factorial design; the audit removes the most immediate single-seed explanation but does not establish seed-averaged statistics.

4.4 Late-round degradation is Stage-B-led with frequent coupling

The alternating-round degradation audit analyzes 11 runs and 38 round-to-round transitions. Nine transitions worsen the Stage C closed-loop head RMSE. All 9 of these worsening transitions include a rebound in Stage B $\log K$ RMSE, while 6 also include a rebound in Stage A head RMSE. On worsening transitions, the correlation between the Stage B $\log K$ -RMSE change and the Stage C head-RMSE change is 0.784.

This result argues against a simplistic explanation in which the workflow failure is only a Stage A head-fitting problem. Late degradation is better described as Stage-B-led with frequent Stage A/B coupling near the stopping point.

4.5 Direct-map sensitivity shows linearized local observability with increasing conditioning burden

The finite-difference sparse-head Jacobian diagnostic gives full relative rank for the tested 3/5/10-mode direct maps at $n_{\text{obs}} = 200$. The Jacobian condition number increases with mode count (Table 1, Figure 3).

This result does not prove global identifiability. Full column rank of the Jacobian at a reference point implies linearized local observability—that is, small perturbations in all coefficient directions are detectable by the sparse-head map at the linearization point. Global identifiability, uniqueness, and optimization convergence are not

Table 1 Jacobian and Gauss-Newton condition numbers for 3/5/10-mode direct FDM-KLE maps at $n_{\text{obs}} = 200$.

Mode count	Rank	$\kappa(\mathbf{J})$	$\kappa(\mathbf{H}_{\text{GN}})$
3	3/3	10.6	112
5	5/5	35.6	1.27×10^3
10	10/10	54.3	2.95×10^3

implied. The tested observation layout shows linearized local observability through 10 modes. Therefore, higher-mode hybrid failure should not be described as proven absence of local information. It is more consistent with workflow damage and increasing conditioning burden.

The same 3-mode diagnostic was then repeated at the three compact-matrix display truth realizations (1, 2, and 3) with $n_{\text{obs}} = 200$. All three cases remain full rank (3/3). Their Jacobian condition numbers are 10.8, 7.53, and 11.4, respectively, with Gauss-Newton condition numbers 117, 56.7, and 131. This truth sweep is important because truth 2, one of the severe PINN-hybrid failure cases, is not the worst conditioned direct map; it is the best conditioned of the three in this local diagnostic. Thus, the compact truth results strengthen the workflow-limited interpretation: the tested hybrid failures do not coincide with loss of local direct-map observability at the true coefficients.

To check whether the full-rank finding is an artifact of the observation density, we repeated the truth-point FIM diagnostic for the same three display truths across four observation layouts ($n_{\text{obs}} = 50, 100, 200, 400$) at $\sigma = 0.005$. All 12 cases remain full rank (3/3). Truth 2 retains the best conditioning across all densities, with Jacobian condition numbers 6.4, 6.9, 7.5, and 7.8 at $n_{\text{obs}} = 50, 100, 200, 400$, respectively (compared with 9.1–11.0 for truth 1 and 10.3–11.8 for truth 3). The FIM rank is unaffected by reducing observations from 400 to 50. This observation-density sweep (Figure 2) supports the workflow-limited interpretation: rank deficiency is not observed at any tested density, yet the staged workflow fails to extract that information into recovered coefficients at $n_{\text{obs}} = 200$.

We further evaluated local sensitivity at the selected hybrid-recovered coefficients for both hard-failure truths (2 and 3), using the six targeted seed-audit cases each ($\sigma = 0.005/0.02$,

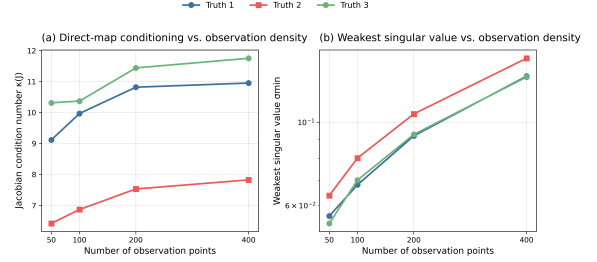


Fig. 2 Jacobian condition number and weakest singular value across observation densities for the three compact-matrix truth realizations. All Jacobians remain full rank (3/3) from 50 to 400 observation points. Truth 2 retains the best conditioning at all densities.

seeds 202/303/404, $n_{\text{obs}} = 200$). Across the twelve paired cases, recovered- ξ distances range from 1.54 to 4.59, showing that the hybrid workflow places the coefficients far from the truth. Nevertheless, recovered-point local direct-map conditioning remains comparable to true-point conditioning. The recovered/true Jacobian condition-number ratio $\kappa(\mathbf{J}_{\text{rec}})/\kappa(\mathbf{J}_{\text{true}})$ ranges from 0.919 to 1.010 with median 0.953, and all twelve recovered-point Jacobians remain full rank (3/3). The recovered/true weakest-singular-value ratio ranges from 0.930 to 1.022 with median 0.979.

For truth 2, recovered- ξ distances are between 4.13 and 4.59, with condition-number ratios 0.935–1.010. For truth 3, recovered- ξ distances are between 1.54 and 2.57, with condition-number ratios 0.919–0.959. In both hard-failure truths, the hybrid workflow does not appear to fail by moving into a locally unobservable region of the direct FDM-KLE map. It moves to wrong coefficient values that still have locally observable sparse-head sensitivity. This replicated finding across two truths strengthens the workflow-limited interpretation: the failure mechanism is not loss of local direct-map observability at the recovered parameters (Figure 4).

The close agreement of condition numbers between true and recovered points (ratio 0.919–1.010) indicates that the local sensitivity spectra at the true and recovered points are comparable. This observation supports the side-by-side comparison of true-point and recovered-point local diagnostics, but similar condition numbers do not, by themselves, imply that the two points lie in the same attraction basin of the optimization landscape.

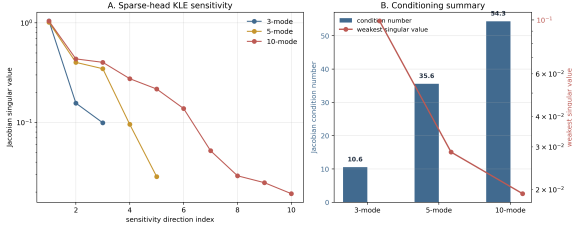


Fig. 3 Finite-difference sparse-head Jacobian spectrum diagnostics. Panel A shows the singular value spectrum of the observation Jacobian $\mathbf{J}$ for 3-, 5-, and 10-mode KLE cases. Panel B shows the Jacobian condition number $\kappa(\mathbf{J})$ (left axis, bars) and the weakest singular value (right axis, line) versus mode count ($n_{\text{obs}} = 200$). The Fisher information and Gauss-Newton Hessian spectra are determined from $\mathbf{J}^T \mathbf{J}$ and $2\mathbf{J}^T \mathbf{J}$ respectively and are not plotted separately.

4.6 Mode bridge separates workflow damage from structural claims

Mode-bridge diagnostics compare hybrid recovery with fixed-head coefficient recovery. At 3 modes, workflow damage is visible but still tolerable in the representative bridge case: hybrid logK RMSE is 0.125, while perfect-head recovery gives logK RMSE 1.72×10^{-4} . At 5 and 10 modes, the hybrid workflow fails with logK RMSE 1.219 and 1.426, respectively, while perfect-head recovery remains near exact (7.42×10^{-4} and 1.35×10^{-3}).

The conclusion is that 5/10-mode failure in the current hybrid workflow is not evidence of structural impossibility. It is a workflow-limited failure under the tested staged approximation and recovery procedure.

5 Discussion

5.1 Workflow-limited failure is distinct from non-identifiability

The main result is the separation between local information content and workflow recovery. The direct FDM-KLE baseline and the sparse-head Jacobian diagnostic show that low-dimensional coefficient information can be present in the direct inverse map. The PINN-hybrid workflow can still fail to recover that information. This is not a semantic distinction: it determines whether one should conclude that the observations are intrinsically insufficient or that the tested workflow

fails to reliably preserve recoverable coefficient information.

For the present benchmark, the moderate-noise 3-mode failure is best described as workflow-limited. This phrase should be read narrowly. It means workflow-limited relative to the tested PINN-hybrid design and the direct FDM-KLE baseline in the same reduced KLE space. It does not imply that all PINN-based groundwater inverse formulations fail.

For clarity, we distinguish four failure categories relevant to this benchmark: (i) *information-limited* failure, where the observations do not contain coefficient-level information in any reduced map; (ii) *optimization-limited* failure, where information is present in the direct map but the optimizer cannot reach it; (iii) *architecture-limited* failure, where the neural representation cannot approximate the required function space; and (iv) *stage-coupling* failure, where errors in one training stage propagate irreversibly into subsequent stages (these categories are not mutually exclusive). The present workflow-limited finding combines elements of (ii) and (iv): the direct map contains recoverable information, but the staged A/B optimization pathway does not reliably transmit it into recovered coefficients. We note that establishing the precise causal chain from the stage-coupling mechanism (iv) to the observed recovery gap would require tracing the optimization trajectory from initial to recovered parameters; the present diagnostic framework identifies the gap and its stage-of-origin, but does not fully resolve the mechanistic link between them.

5.2 Relationship to classical inverse modeling

The direct FDM-KLE baseline plays a role similar to classical deterministic parameter estimation in a reduced parameter space [5, 8, 26]. It is not a full comparison against PEST, EnKF, EnRML, MCMC, or geostatistical inversion software [27–29]. However, it is strong enough for the specific diagnostic claim made here: if direct FDM-KLE least squares recovers the same coefficients from the same observations while PINN-hybrid fails, then the failure cannot be assigned solely to low-dimensional information absence.

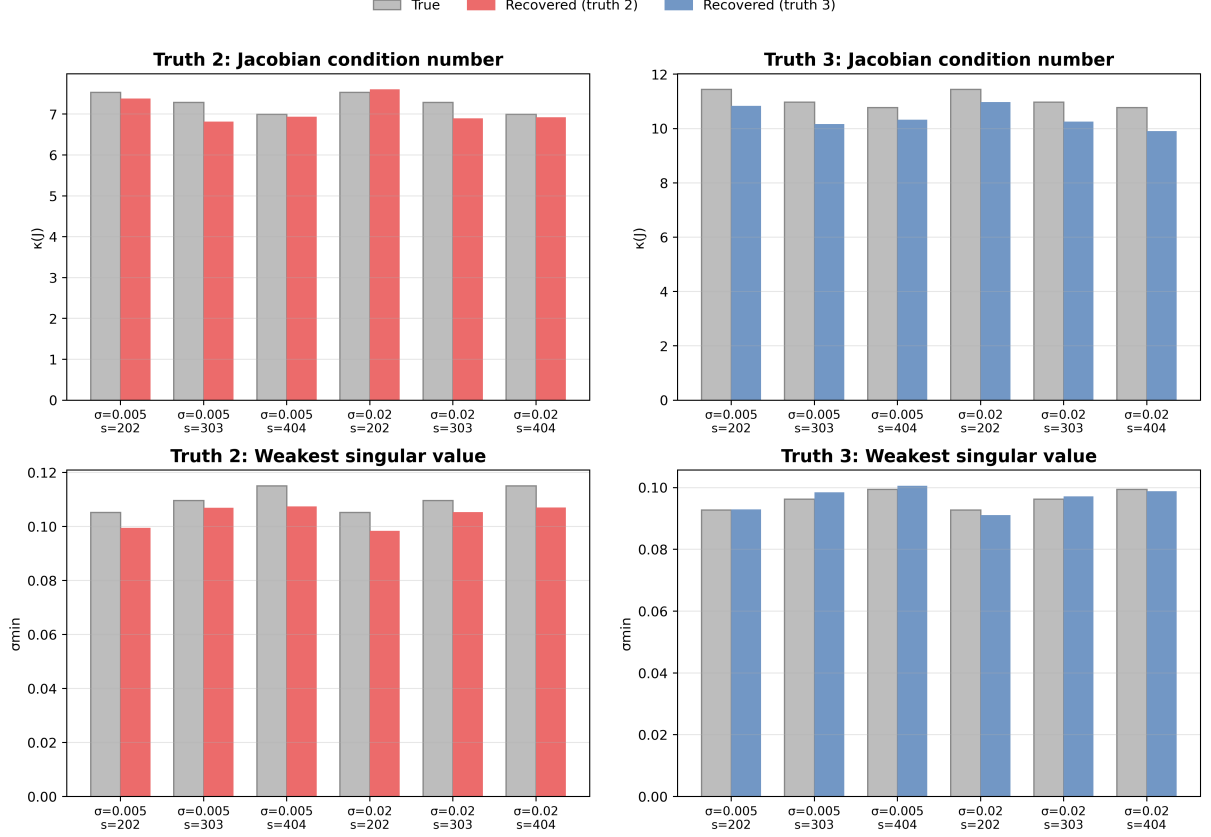


Fig. 4 Recovered-point versus true-point local conditioning for two hard-failure truths (2 and 3). Panels A/C show Jacobian condition numbers; panels B/D show weakest singular values. Each panel pairs true and recovered values across six seed-audit cases per truth. The dashed line marks equal conditioning. All recovered points remain full rank (3/3) despite coefficient distances of 1.54–4.59 from truth.

A broader comparison with PEST-style regularized inversion, adjoint least-squares, or ensemble methods would be valuable, but it is not required to establish the workflow-limited finding. Those methods would address a wider question: which inverse family is most accurate or efficient? The present paper addresses a narrower diagnostic question: whether the tested PINN-hybrid workflow loses recoverable information. The comparison is diagnostic rather than competitive: we test whether recoverable coefficient information is accessible to a direct solver in the same reduced space, not whether FDM-KLE is a superior inversion method for practical applications.

5.3 What the FIM and Hessian diagnostics do and do not prove

The Jacobian, Fisher-information, and Gauss-Newton diagnostics are local and linearization-based [9, 10]. A full-rank Jacobian at a reference point does not prove global uniqueness, rule out multimodality, or quantify posterior uncertainty under nonlinear effects. It only shows that small perturbations in the tested coefficient directions are locally visible to the sparse-head observation map. Conversely, increasing condition number indicates increasing sensitivity burden, not categorical failure.

This local interpretation is sufficient for the paper’s logic. The FIM diagnostic is used to reject an overly strong structural claim: the 5/10-mode hybrid failures should not be reported as if the direct observation map had no local coefficient

information. They should instead be presented as workflow failures under a more difficult conditioning regime.

The recovered-point FIM analysis (Section 4.5) adds an important nuance to this local interpretation. When the same Jacobian diagnostic is evaluated at the hybrid-recovered coefficients rather than at the true coefficients, the direct-map remains full rank with comparable conditioning (recovered/true condition-number ratio: 0.919–1.010, median 0.953, across twelve paired cases from two hard-failure truths). This means that the hybrid workflow does not fail by drifting into a locally unobservable region of parameter space. It reaches coefficient values that remain locally observable under the direct linearized map.

This finding has two implications. First, it further strengthens the workflow-limited interpretation: the failure mechanism is not loss of local direct-map observability at the recovered parameters. The local sensitivity spectra at true and recovered points are comparable; what is lost is the optimization pathway that connects the data to the correct coefficient values. Second, it illustrates an important limitation of local observability diagnostics: a full-rank Jacobian at a point does not guarantee that an iterative method can reach that neighborhood from a distant starting point, nor does it guarantee that a different optimization landscape (here, the PINN-hybrid loss surface) shares the same favorable geometry as the direct least-squares landscape.

The distinction between local observability at a point and recoverability by a specific optimization procedure is therefore not merely semantic. A full-rank FIM at the true coefficients tells us that the observations contain low-dimensional information; a full-rank FIM at the recovered coefficients tells us that the optimization has not collapsed into an unobservable region; but neither guarantees that the PINN-hybrid training trajectory can connect the starting point to the correct solution. This three-way separation—information content, local geometry, and optimization pathway—is the essential diagnostic structure of the workflow-limited failure.

5.4 Transfer to field applications

The synthetic results support a transfer in logic rather than a transfer in numerical thresholds. The exact noise and observation-density boundaries are specific to the steady unit-square benchmark. Real aquifers may have pumping, recharge, transient responses, geological boundaries, irregular observation layouts, and independent measurements that change identifiability [12, 13]. The limitation also follows the broader modeling principle that controlled models can be useful diagnostic tools without being validated as direct field surrogates [30].

What is likely to transfer is the evaluation principle. Head fit should not be equated with conductivity recovery. Reduced parameterizations should be matched to the information content of the data. Direct deterministic baselines and closed-loop validation should be reported before claiming PINN-based conductivity inversion success. When sparse head-only data are the dominant information source, unconstrained conductivity maps should be treated with particular caution.

5.5 Limitations and next steps

The current study is synthetic, steady, source-free, and uses regular boundary conditions. The main noise matrix now includes a compact eight-truth FDM check, but seed coverage is still concentrated in targeted audit cases rather than a full factorial ensemble. The FIM diagnostic has been repeated across the compact truth realizations for $n_{\text{obs}} = 200$, at recovered points for two hard-failure truths (2 and 3, twelve paired cases), and across four observation densities (50–400), but should still be extended to truth/recovered-point combinations across densities. The direct baseline is deterministic FDM-KLE least squares, not a complete hydrogeologic inversion ecosystem [8, 26].

The most important next extension is not to make the benchmark broader in every possible direction, but to make the current diagnostic claim statistically firmer. The 3-mode noise/density matrix has been expanded to eight truth realizations for the direct FDM-KLE baseline (Section 4.3), and the Jacobian diagnostic

has been extended across observation layouts (50–400, Section 4.5). A natural next stage would be to repeat the matched hybrid comparison across the same expanded truth set and to extend the recovered-point diagnostic across observation densities. More general hydraulic excitations, transient cases, and Bayesian posterior analysis are valuable later extensions.

The current study deliberately isolates the staged A/B workflow because it represents one practical way to separate head approximation from coefficient recovery in PINN-KLE inverse studies [16, 17]. A joint end-to-end optimization that updates h and ξ simultaneously may alleviate the stage-coupling failure mode (iv), but would introduce its own convergence challenges in the combined landscape; evaluating such alternatives is beyond the present diagnostic scope and would be a valuable extension.

5.6 Evidence-type labeling for identifiability claims

The reduced-parameterization experiments demonstrate that identifiability claims must be labeled by evidence type. Scalar effective conductivity is structurally unidentifiable under the tested steady source-free setup because changing a homogeneous scalar conductivity does not change the head solution. Homogeneous axis-aligned anisotropy is also structurally unidentifiable under the same left-right loading. A two-zone contrast becomes meaningful when an absolute scale prior is imposed. Rotated anisotropy is numerically recoverable in the tested homogeneous-tensor setting under multistart optimization, but local Hessian analysis reveals shallow valleys.

These cases should not be collapsed into a single vague “identifiability” scale. They represent analytic structural arguments, numerical invariance tests, optimization recoverability, local-curvature uncertainty, and workflow-limited diagnostics. The PINN-KLE diagnostic framework in this paper addresses the last two categories; it does not replace structural analysis or geometric invariance arguments.

6 Conclusions

This paper presents a controlled numerical diagnostic study of PINN-KLE groundwater conductivity inversion under sparse steady head observations. The main conclusions are:

1. In the tested staged PINN-KLE workflow, the head-observation loss provides no direct gradient pathway to KLE coefficients; updates are mediated through the PDE-residual branch in the implemented staged workflow.
2. No-truth round selection can identify the best available alternating round, but selecting the best round is not equivalent to conductivity recovery.
3. Direct FDM-KLE least-squares passes 24/24 cases across an expanded set of eight truth realizations ($\sigma_{\log K} = 0.17\text{--}0.80$, three noise levels), showing that the sparse direct inverse problem still contains recoverable low-dimensional information under the tested 3-mode parameterization. In the matched three-truth seed audit, the PINN-hybrid workflow weak-passes only 3/18 conductivity-recovery cases. This gap makes the moderate-noise hybrid failure workflow-limited in this setting.
4. Late-round hybrid degradation is Stage-B-led with frequent Stage A/B coupling.
5. Sparse-head Jacobian/FIM/Gauss-Newton diagnostics show linearized local observability for the direct map across observation densities from 50 to 400, with increasing conditioning burden as KLE mode count increases.
6. Recovery-point local conditioning remains comparable to true-point conditioning (condition-number ratio 0.919–1.010, median 0.953, across twelve paired cases), showing that the observed hybrid recovery failures are not caused by the workflow moving the coefficients into a locally unobservable region.

The broader recommendation is methodological: PINN-based groundwater inverse studies should distinguish head fitting, linearized local observability, deterministic baseline recoverability, and closed-loop conductivity recovery. Without this separation, a successful PINN head fit can be mistaken for a successful hydraulic-conductivity inversion. This study focuses on diagnostic information accessibility in a controlled

benchmark, not on proposing a field-ready inverse solver.

A Training and baseline configuration

This appendix summarizes the computational settings used in the reported experiments. All values are extracted from the YAML configuration files and source code in the working repository. The pilot experiments (Section 4.3, pilot aggregate) use high-budget settings; the CG compact multi-truth matrix and seed audit use medium-budget settings to keep the experimental matrix computationally tractable. Both configurations are documented below.

A.1 PINN-hybrid workflow

Table 2 Neural network and domain configuration.

Setting	Value
Neural network depth	5 hidden layers
Neural network width	128 neurons per layer
Activation function	tanh
Weight initialization	Xavier
Hard Dirichlet BC	Yes (left/right enforced analytically)
Fourier features	8 frequencies, max 8.0, linear spacing
KLE covariance kernel	Exponential, $\sigma^2 = 1.0$, $l_x = l_y = 0.2$
FDM reference grid	51×51

A.2 Stage-wise settings: high-budget (pilot) vs. medium-budget (CG compact)

A.3 Direct FDM-KLE baseline

The direct baseline optimizes the same low-dimensional KLE coefficients as the PINN-hybrid workflow, but evaluates each candidate through the FDM forward solver and minimizes the sparse-head residual directly. It is included as a diagnostic comparator rather than as a claim of best-practice hydrogeologic inversion.

Table 3 Common optimizer and iteration settings.

Setting	High-budget (pilot)	Medium-budget (CG compact)
Stage A iterations/round	500	500
Stage B iterations/round	1000	1000
PDE collocation points	2000	1000
BC sampling points	400	200
Alternating rounds (max)	6–8	5
Closed-loop validation grid	201×201	101×101
Observation seeds	202, 303, 404	202, 303, 404

Table 4 Stage A (head fitting, ξ frozen). Common to both budgets.

Setting	Value
Optimizer	Adam
Learning rate	5×10^{-4}
Loss weights: PDE / BC / Data	1.0 / 10–20 / 100.0
Weight decay / Gradient clip	10^{-5} / 1.0
LR scheduler	Equal to iteration count (disabled)

Table 5 Stage B (coefficient recovery, h frozen). Common to both budgets.

Setting	Value
Optimizer	Adam
Learning rate	10^{-2}
Coefficient initialization	From previous round (round 1: zero)
h smoothing σ	0.0 (no Gaussian smoothing)
logK clamping	± 3.0

A.4 Jacobian/FIM diagnostic

The local sensitivity diagnostic uses central finite differences of the sparse-head observation map with respect to KLE coefficients. The reported FIM and Gauss–Newton spectra are derived from the same Jacobian and are used only as local linearized observability checks.

Table 6 FDM-KLE baseline configuration.

Setting	Value
Optimizer	scipy.optimize.minimize (Powell)
Objective	Sum of squared sparse-head residuals
Forward model	FDM on 81×81 grid
Max iterations	80
Random starts	2
Coefficient bounds	± 3.0 in ξ
Closed-loop validation	FDM on 201×201 grid

Table 7 Jacobian/FIM diagnostic configuration.

Setting	Value
Finite-difference step	$\delta = 10^{-3}$ (absolute perturbation)
Reference points	(a) true ξ , (b) hybrid-recovered ξ
Observation densities scanned	50, 100, 200, 400
Noise level for FIM scan	$\sigma = 0.005$

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Statements and Declarations

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Competing interests

The author declares no competing interests.

Ethics approval

Not applicable. This study is a purely computational investigation using synthetic data and does not involve human participants or animals.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The machine-readable summary tables and meta-data supporting the reported numerical results are supplied as Online Resource 2 for review, including the FDM-KLE baseline metrics, PINN-hybrid recovery statistics, Jacobian/FIM diagnostics, and figure-level summary data. The raw NumPy arrays for truth realizations, synthetic observation sets, and FDM reference solutions are retained in the project output tree and are available from the author upon reasonable request during review. A public archive containing the review data package and the raw numerical arrays will be deposited in Zenodo or HydroShare before publication, and the persistent identifier will be added to the final article if the manuscript is accepted.

Code availability

The Python source code implementing the finite-difference forward solver, PINN-KLE training workflow, FDM-KLE least-squares baseline, Jacobian/FIM diagnostic scripts, and figure-generation utilities is summarized in Online Resource 2 for review and is publicly available at <https://github.com/LancelYang/pinn-kle-groundwater-diagnostics>. The training configurations (YAML) are self-documented in Appendix A, and the exact command-line invocations for each experiment are recorded in `runner_command.json` files within each experiment output directory. The computational environment uses Python 3.11+ with PyTorch and NumPy/SciPy as the primary dependencies; dependency files are included in Online Resource 2.

Author contribution

F.Y. conceived the study, designed the diagnostic framework, implemented the experiments, and wrote the manuscript.

Use of AI-assisted tools

During the preparation of this work, the author used a coding assistant powered by GLM-series large language models for code debugging, code refactoring, and language polishing. The tool

was not used for data analysis, scientific interpretation, or the generation of conclusions. The author reviewed all AI-assisted outputs and takes full responsibility for the scientific content and reported results.

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